

Rolling Stock Scheduling

Short Description

As a railway operator DSB faces the challenge of scheduling train operations while balancing infrastructure limitations, passenger demand, and resource efficiency. The planning process involves complex decisions, such as allocating rolling stock, managing capacity, and adapting to varying track conditions.

Together with the infrastructure manager, Bane Danmark, DSB agrees on a timetable that respects the limitations of the railway tracks. Once the timetable is finalized, DSB must create a rolling stock and station schedule that aligns with the timetable while considering passenger flow and station capacity.

Currently, this scheduling process is done manually, with only minimal system support features, such as shortcuts and basic list validations. One of the key goals of rolling stock scheduling is to ensure that the planned trains have sufficient capacity to meet passenger demand. This involves selecting the right train formations, i.e., determining the number of units needed for a train on a specific route or sub-route.

Besides train formations, also the rotation problem should be solved. The rotation problem is to build the plan for every train unit (train set, locomotive and coaches). Part of the rotation problem is to ensure flow conservation, i.e., the number of units arriving to a station must equal the number of departing units.

To make the most of the limited capacity on the tracks and ensure high customer service, DSB plans to couple and decouple rolling stock during operations based on the timetable. For example, a train traveling from København to Jylland with two units can be decoupled in Fredericia, allowing one unit to continue to Esbjerg and the other to Aarhus. This approach utilizes the limited capacity over Storebælt while ensuring that passengers traveling to Esbjerg and Aarhus can remain in their seats for the entire journey. However, planning these “Through Coaches” adds significant complexity to the scheduling process.

Another challenge in rolling stock scheduling is the variation in track electrification. For instance, while some tracks are electrified, others, like the section from Fredericia to Aarhus, are not yet electrified. This requires careful selection of the appropriate rolling stock type for each route. Additionally, different rolling stock types have varying maintenance costs, and the availability of specific types is limited.

Inputs to the rolling stock scheduling are the infrastructure (data in XML), timetable (data in XML), the passenger demand (data in XML), and rules and costs (data in Excel or PNG).

Through Coaches

Through-going coaches (GGV = **G**ennemgående **v**ogne) are a central part of DSB's commercial offering. They allow passengers to travel directly between København and several cities in Jylland (e.g., Aarhus, Aalborg, Struer, Sønderborg) while reducing the number of trains crossing Storebælt, where track capacity is limited.

To make the most of the limited capacity and maintain high customer service, DSB plans to couple and decouple rolling stock during operations based on the timetable. For example, a train traveling from København to Jylland with two units can be split in Fredericia, with one unit continuing to Sønderborg and the other to Aarhus. This approach utilizes the limited capacity over Storebælt while ensuring that passengers traveling to Sønderborg or Aarhus can remain in their seats for the entire journey. In this case, GGVs exist between København and Sønderborg, as well as between København and Aarhus. However, planning these GGVs adds significant complexity to the scheduling process.

To meet passenger expectations, DSB must ensure that a minimum number of units remain attached to the train throughout the entire journey. This requires a system to track which units have been part of the train since the first trip and which units were added later.

A GGV can also exist between two trips without any coupling or decoupling of units. For example, a train traveling from Aalborg to Aarhus (train no. 4074, AB 20:03 – AR 21:13) and another train traveling from Aarhus to Fredericia (train no. 74, AR 21:20 – FA 22:21) with only one unit. If there is a GGV between trips 4074 and 74, it means the same rolling stock unit operates both trips. In this case, passengers can be informed that it is possible to make a seat reservation for the entire journey from Aalborg to Fredericia.

While it may be possible to create a feasible or even optimal rolling stock schedule without GGVs, including GGVs significantly enhances customer service by ensuring seamless travel and reducing the need for passengers to change seats or trains.

Important things from DBS side:

- Cover all trains in the timetable (note: not M-trains)
- Limit costs
- Cyclic rolling stock schedule
- High customer service (limit standing passengers and incorporating **GGV**)